NYC Citi Bike Demand Forecasting & Optimization

A reproducible applied mathematics and operations research study using official 2025 completed-ride records

Abstract

This study asks whether next-hour departure and arrival forecasts can improve simulated service when used to allocate a fixed rebalancing budget. Twelve official monthly archives contain 45,771,102 records, of which 45,588,101 survive documented cleaning. Thirty stations are selected using January-August data only. Last-hour, seasonal-naive, Poisson and XGBoost models are evaluated chronologically on November-December. Integer optimization and distance-greedy transfers are compared with no rebalancing under identical restrictions.

XGBoost achieves departure MAE 3.418 and arrival MAE 3.398 retained rides per station-hour. In the hourly count scenario, MILP raises the service proxy by 1.389 percentage points relative to no rebalancing, while requiring transport. A separate paired-ride counterfactual replay suppresses returns after rejected departures and verifies fleet accounting. Twenty-one sensitivity/ablation scenarios and paired daily bootstrap intervals provide evidence within the assumed models.

These are not historical stockout estimates. Completed trips censor unobserved demand, the original stress baseline assumes stocks and capacities (new evidence is in Section 8), and archived-count zeros do not establish station operation. The measured benefit is conditional on scenario assumptions, not an identified operational or financial impact.

Research contributions

The contributions are an audited full-year data pipeline; leakage-tested count forecasting; an explicit integer formulation; fair count and paired-event simulations; and generated, reproducible results. Negative comparisons are retained: a Poisson model does not automatically outperform a last-hour predictor, and greater service need not mean less transport.

Reproduction: README.md, requirements-lock.txt, scripts/official_pipeline.sh and reports/validation/official_run_evidence.json. All numbers in this report are read from saved outputs; no target effect was specified in advance.

1. Data, cleaning and observation model

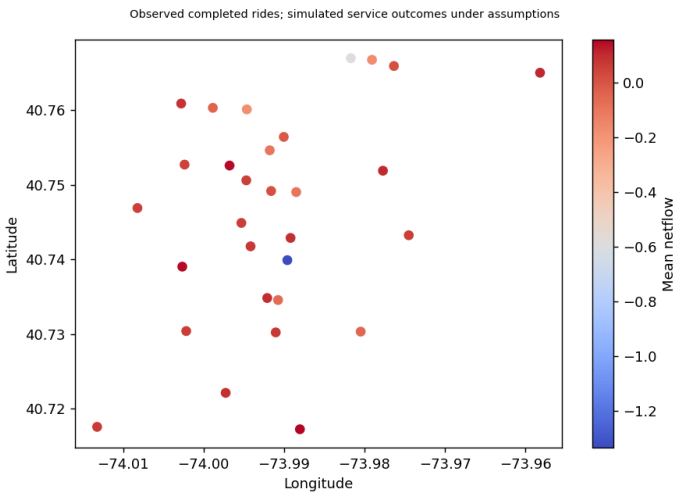
Source: Citi Bike System Data and https://s3.amazonaws.com/tripdata/. The 12 NYC ZIPs contain 53 CSV members and 8,934,786,928 downloaded bytes. HTTPS, ZIP CRC and local SHA256 checks are retained. Raw files are immutable; ZIP and extracted duplicates are not double counted. String station IDs preserve their representation.

Item	Value
Raw rows	45771102
Retained rides	45588101
Selected stations	30
Station-hour grid	262800
Operating-unknown hours	19634

Cleaning rejects missing required IDs, invalid or unresolved timestamps, nonpositive/over-180-minute durations, and implausible coordinates. Ride-ID deduplication is global and disk-backed. Each removal reason and file range is audited. Section 9 adds a full raw-data cutoff sensitivity through 360 minutes. Endpoint flows crossing the study boundary are retained and independently reconciled to cleaned rides.

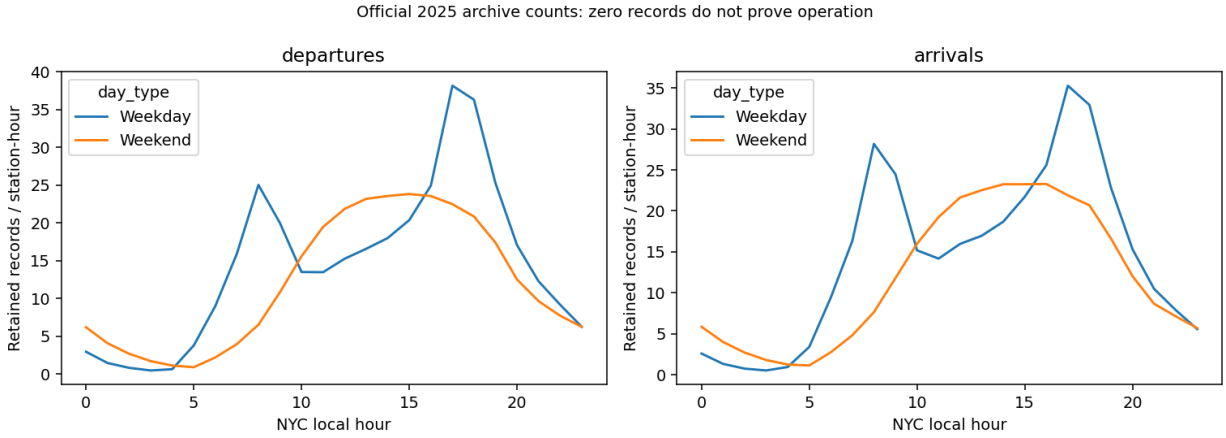
Selection uses January-August activity, at least eight active months, a volume threshold, a Manhattan-area rectangle and conservative coordinate-drift summaries. It does not use validation/test popularity. The rectangle is not an official borough polygon; chunk/month coordinate summaries are not verified relocation histories.

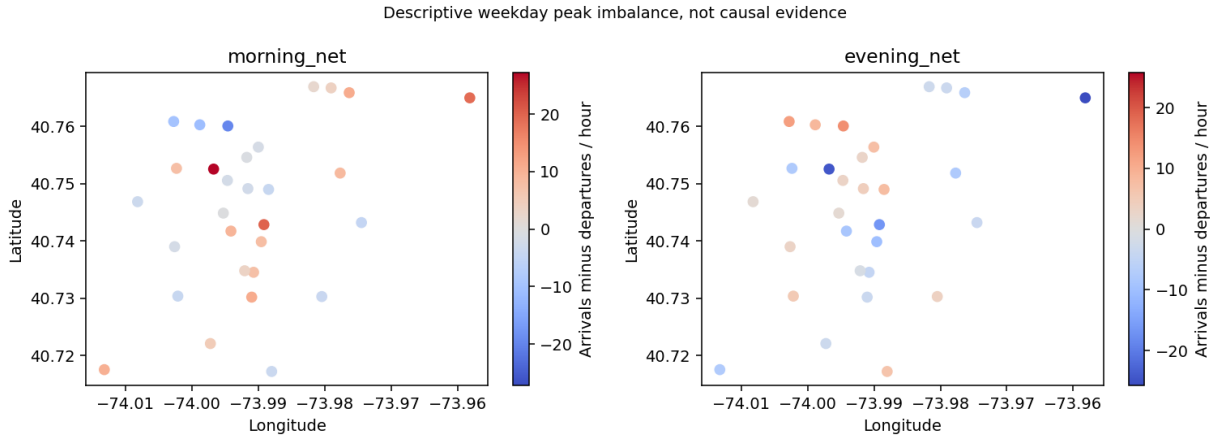
The audited panel keeps operating-unknown hours missing. The explicit recorded_event_counts experiment instead uses retained archive counts in a temporary modeling copy, where zero means no retained record, not proven operation or zero latent demand. Labels and lags use the same estimand. Naive DST folds/gaps are excluded rather than assigned an invented offset. UTC hours remain continuous; local March/November days have 23/25 hours.



2. Exploratory evidence and hypotheses

Daily/hourly trends, weekday/weekend profiles, local-hour heatmaps, station maps and netflow summaries describe the selected network. Peaks motivate studying transport, but do not establish weather effects, closures or real lost demand. The following comparisons use retained counts; active-hour conditional means in other audited-panel plots are distinguished in the data dictionary.





station_name	morning_net	evening_net
W 43 St & 10 Ave	-19.6948	14.1162
11 Ave & W 41 St	-10.8531	8.6245
12 Ave & W 40 St	-9.8314	12.2337

Negative netflow means departures exceed arrivals; opposite-signed stations are potential complements subject to distance and available stock. The table identifies morning outflow candidates without claiming they actually ran empty. Afternoon reversals and spatial imbalance guide the hypothesis tested by the optimizer. These descriptive full-year figures were not used to retune the held-out test models.

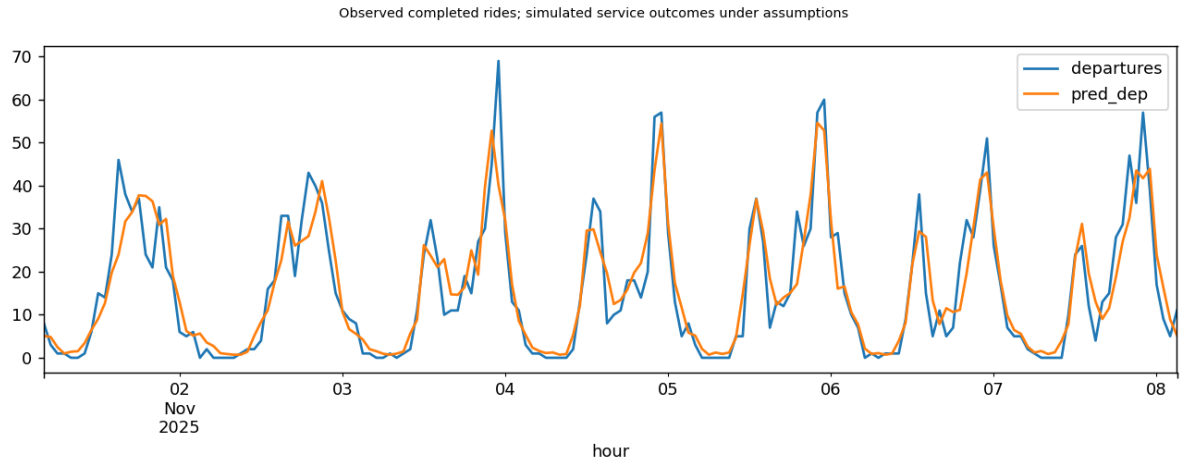
3. Forecasting protocol and out-of-sample errors

Train: January-August. Validation: September-October. Test: November-December. Separate target models predict departures and arrivals at the start of each hour. Rolling one-step evaluation may use earlier observed test hours; it does not forecast two months once from October. Both final target models are selected on validation and refit on train+validation, with no test-based hyperparameter selection.

Features include local hour/day/month/weekend/federal holiday, cyclic hour/week terms, station one-hot, lags 1/24/168 and shifted 24/168-hour means/stds. Imputation/scaling/encoding are fitted on training only. Elapsed-hour lags across DST need not equal the same wall-clock hour. The baseline omits weather; Section 9 adds verified 24-hour-lead archived forecasts. Calendar-only and lag-based inputs do not use future target-hour weather or rides.

model	target	mae	rmse	wape
last_hour	departures	4.4758	7.1345	0.4520
seasonal_naive	departures	5.7187	9.1542	0.5775
poisson	departures	4.9315	6.6522	0.4980
xgboost	departures	3.4177	5.0567	0.3451
last_hour	arrivals	4.4091	7.2007	0.4487
seasonal_naive	arrivals	5.7217	9.2423	0.5823
poisson	arrivals	4.8384	6.6387	0.4924
xgboost	arrivals	3.3982	5.1112	0.3458

Units are retained rides per station-hour for MAE/RMSE, and a ratio for WAPE. Undefined zero-denominator WAPE is missing, not zero. Test rows number 43,950 (30 stations x 1,465 hours). Errors are stratified by peak, weekday/weekend, demand group and test month in forecast_metrics.csv. Training errors are in-sample diagnostics. Poisson here is a regularized log-link count regression, not an inferential unpenalized GLM.



4. Integer rebalancing formulation

Let I_i be current simulated stock, C_i assumed capacity, D_i/A_i predicted departures/arrivals, and d_ij Haversine distance. Integer x_ij moves bikes from i to j at the hour boundary; continuous u_i/v_i penalize forecast-end departures from safety bounds L_i/U_i. The objective is a proxy for balance and transport, not exact within-hour loss.

```
min w_u sum(u_i) + w_v sum(v_i) + w_d sum(d_ij x_ij)
Ipost_i = I_i + sum_j x_ji - sum_j x_ij
E_i = Ipost_i + A_i - D_i
u_i >= L_i - E_i; v_i >= E_i - U_i; u_i,v_i >= 0
0 <= Ipost_i <= C_i; sum_ij x_ij <= B; x_ij integer >= 0
```

Only supply-to-deficit edges within radius exist, excluding self-loops and transshipment. Define midpoint M=(L+U)/2, donor supply s_i=min(I_i,max(0,floor(I_i+A_i-D_i-M_i))) and receiving bound r_i=min(C_i-I_i,max(0,ceil(M_i-I_i-A_i+D_i))). Outbound sums cannot exceed s_i; inbound sums cannot exceed r_i. These extra restrictions are shared by Greedy and MILP and imply capacity feasibility. Zero movement plus unrestricted slack is feasible for valid inputs.

SciPy milp/HiGHS uses a five-second limit and zero requested relative MIP gap. Only successful solves are called optimal. A failure keeps the solver status and uses a logged Greedy fallback. All official baseline hours solved without fallback. Each move is checked for nonnegative integrality, stock/capacity, budget and permitted edges. Mean official MILP runtime was 7.882 ms on this instance; this is not an operational latency guarantee.

Hand example: I=[3,27,8], C=[30,30,30], D=[15,5,17], A=[2,3,3]. With safety fractions .3/.7, weights 5/5/.1, budget 3 and one-km supply edges, no-move end stocks are [-10,25,-6], proxy cost 190. Moving three bikes reduces it to 160.3. Automated exhaustive enumeration verifies the same optimum and constraints.

Instant transfers ignore trucks, travel delays and driver limits. Distance is not a road route. Capacity=25, budget=12 bikes/hour, radius=3 km and initial fraction=.5 are explicit baseline assumptions. Their consequences are explored rather than asserted as historical facts.

5. Simulation protocols and measured outcomes

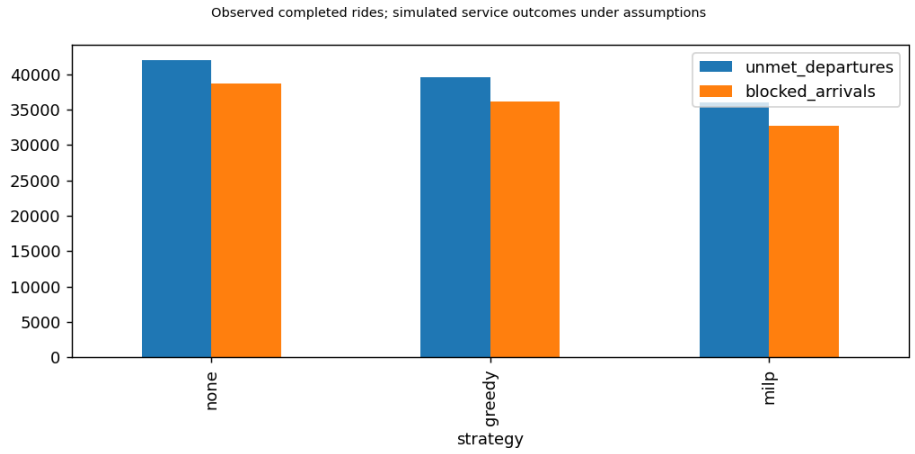
The count scenario first applies each policy then uses identical seeded random departure/arrival order at every station-hour. All policies share scenario requests, capacities, initial stocks, radius and budget. Common sample hashes are checked. Arrivals are exogenous in this model; station balance is verified, but it is not called causal fleet conservation.

strategy	unmet_departures	blocked_arrivals	bikes_moved	bike_km	proxy_service_rate
none	42028	38689	0	0.0000	0.9034
greedy	39510	36167	17259	8,590.2028	0.9092
milp	35982	32639	16442	15,846.8791	0.9173

The paired-event upgrade independently replays 800,854 cleaned rides across the same 1,465-hour test horizon. Rejected departures suppress their own returns. In-transit bikes, initial underway trips and external boundary inflows/outflows are explicit. Full fleet balance is asserted after each processed event. Blocked returns remain held off dock through the horizon; no rerouting is invented.

strategy	unmet_departures	blocked_arrivals	bikes_moved	bike_km	proxy_service_rate
greedy	40827	31632	17181	8,580.9444	0.9062
milp	37832	29174	16267	15,525.9291	0.9131
none	42891	33379	0	0.0000	0.9015

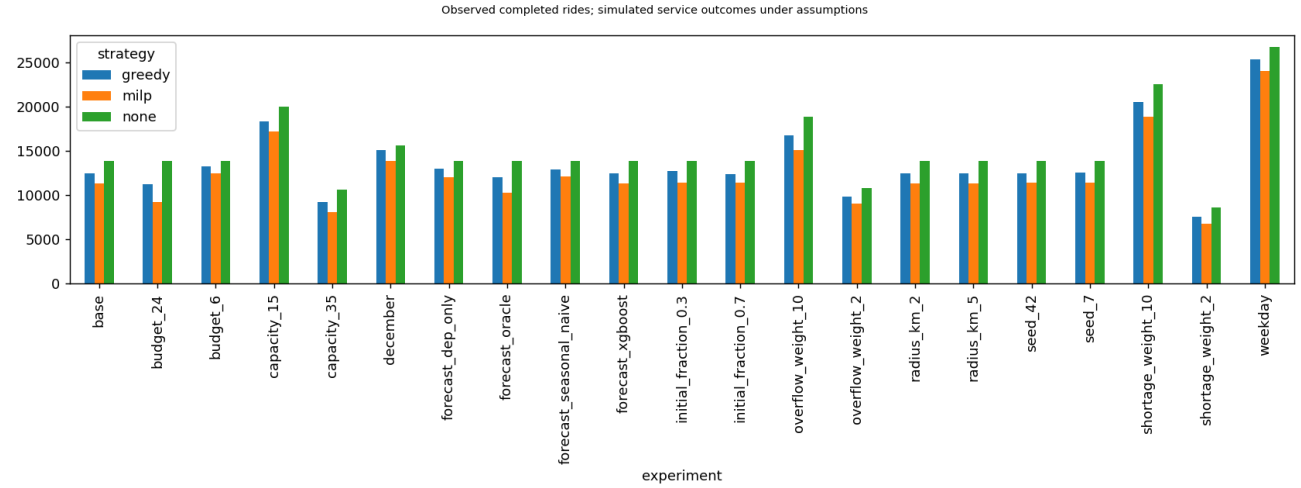
The two rows of evidence are different hypothetical models, not interchangeable estimates of reality. Policy comparison must include service and transport together: the MILP solution uses more bike-km than short-distance Greedy despite moving fewer bikes. Realized preference objective uses penalty weights 5/5 plus .1 per bike-km; neither that score nor bike-km is a measured financial/truck cost.



6. Robustness, ablations and uncertainty

Twenty-one configurations were saved before execution, producing 63 completed policy runs of 48 hours each. One-factor changes cover capacity 15/25/35, budget 6/12/24, radius 2/3/5 km, starting stock 30/50/70%, penalty weights, seeds and test dates. Every scenario, including unfavorable results, is retained in experiments.csv. Scores with different weights are not directly comparable across scenarios.

Forecast ablations use seasonal naive, departures-only with naive arrivals, fixed XGBoost, and future-count Oracle. Oracle is a non-deployable information comparator, not a guaranteed true-service upper bound because optimization targets a proxy. These are exploratory sensitivities on the test set, not a second untouched evaluation. The baseline grid omits weather; Section 9 reports the added weather ablation.



strategy	baseline	ci_low	ci_high
greedy	none	-459.5227	-344.3844
milp	none	-1,029.9151	-901.7282
milp	greedy	-610.9228	-521.4275

Intervals above are paired local-day bootstrap intervals for mean daily scenario-objective differences. Negative values favor the named policy under that objective. Forecast differences also use paired local-day blocks in forecast_daily_bootstrap.csv. There are 61 local test days; local days may contain 23/25 hours. Repeated seeds and resampling quantify within-scenario variability only. They neither recover censored demand nor remove structural bias from unknown stocks, capacities and behavior.

Comparison files report absolute differences and relative changes with named baselines; relative change is undefined for a zero baseline. Model ranking and scenario results are generated from saved files, not manually edited to obtain a desired improvement.

7. Reproduction, limitations and conclusion

Software evidence: 54 tests passed, including the historical evidence, recursive forecast, stochastic solver and routing extensions. Tests cover ZIP/multiple members, duplicate rides, timestamp/DST/duration rules, train-only selection, cross-boundary count conservation, temporal leakage, output alignment, nonnegative forecasts, hand-case MILP optimum, transfer invariants, forced solver fallback, common scenario paths, archive-count semantics and paired-event fleet balance. Independently reconciled official outputs and three executed notebooks are retained.

Prospective monitoring now runs locally at a 60-second target and in a deployed public GitHub Actions workflow at a five-minute target. A successful run produced an archive on gbfs-history that was independently downloaded and restored to SQLite. Every poll retains unchanged counts, flag-only updates, stale/missing states and failures. Verified full-content daily archives precede retention cleanup. Scheduling is best effort; actual dated receipts, gaps and health are in the operations supplement. This does not backfill 2025 or create months of future observations.

Main limitations: completed rides are censored proxies for requests; retained-archive zeros do not establish operation; filtering and source temporal boundaries affect the estimand; complete continuous authoritative histories are not available for every hour (Section 8 documents acquired evidence). The geography is approximate and coordinate drift compressed. Count arrivals are exogenous; paired blocked-return holding and external supply are hypothetical. Current GBFS does not identify historical stockouts. Section 9 adds directed-road delayed truck service and stochastic three-period MPC; actual operator deployment and verified historical traffic remain outside the demonstrated scope.

Conclusion: on the fixed official archive and specified scenarios, XGBoost improves held-out count error over the reported baselines and optimized rebalancing trades transport for higher simulated service. These are reproducible predictive/scenario findings, not a causal demonstration that actual Citi Bike losses or costs decreased. The next credible step is prospective inventory validation before operational deployment.

Reproduction and references

Install: scripts/install.sh. Core reproduction: scripts/official_pipeline.sh with matching prepared data; otherwise download and prepare first. Run python -m pytest, inspect reports/validation, and execute scripts/package.py for an integrity-checked delivery. Raw archives/large intermediates are excluded from Git; manifests and commands reproduce them. Exact dependencies are in requirements-lock.txt; seeds and configurations are saved with every run.

Citi Bike System Data: https://citibikenyc.com/system-data

Official archives: https://s3.amazonaws.com/tripdata/

GBFS feeds: https://gbfs.citibikenyc.com/gbfs/en/

SciPy milp: https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.milp.html

XGBoost: https://xgboost.readthedocs.io/en/stable/python/python_api.html

Project evidence: outputs/official, reports/validation, reports/optimization_formulation.md

8. Newly acquired 2025 historical evidence

The earlier assumption that no historical evidence had been supplied is now superseded. CityBikes publishes 2025 NYC monthly Parquet change-event archives. Four Internet Archive captures of official station_information and six of station_status provide independent dated official observations. Exact short_name-to-UID identities and coordinates match all 30 study stations; names alone are not used.

Evidence	Count
CityBikes 2025 selected-site observations	2907502
Dated official capacity points	120
Dated official status points	180
Rejected inconsistent count/capacity records	15222
Common fresh-state test hours	260

The source collector suppresses unchanged (bikes, free) pairs; flag-only updates may be lost. Code revisions from before and at the end of 2025 document this behavior, although uninterrupted deployment cannot be independently verified. Archive gaps therefore cannot prove uptime or downtime, and event-sampled zero-bike counts are not time-weighted stockout rates.

Of 19,634 previously operating-unknown station-hours, 3,406 have fresh consistent active point support, but only 2 meet the declared all-12-five-minute-bin criterion. The remaining 19,632 fail that stronger criterion and are not fabricated as verified operating hours. Even full bin support is a sampling statement, not continuous proof.

strategy	unmet_departures	blocked_arrivals	proxy_service_rate
greedy	1865	2837	0.9885
milp	1752	2625	0.9892
none	1932	3029	0.9881

Each eligible hour starts from the same preceding historical observation across policies, at most 15 minutes old. Usable capacity equals reported available bikes plus free docks, excluding unusable dock occupancy; physical slots are retained separately. The 260 one-hour scenarios reset to measured states. They are not a continuous replay, and their selected times/initialization differ from the old capacity-25 stress baseline. The rates must not be compared across those tables as a treatment effect.

Bike-share data by CityBikes contributors and available from https://data.citybik.es . CityBikes contributors. (2026) Bike-share data retrieved from https://data.citybik.es . Historical derivative databases are ODbL-1.0; underlying operator terms apply. Manifests, dated official JSON, code revision links and confidence categories are included. NYC inspection data end in 2020 and the reviewed Comptroller analysis covers 2023, so neither is used as 2025 truth.

9. Executed extensions and revised conclusions

All added analyses are explicitly exploratory because the original test set had already been evaluated. No new untouched holdout is claimed. Full methods and per-run files accompany the report.

target	mean_daily_MAE_difference	ci_low	ci_high
departures	-0.0304	-0.0454	-0.0163
arrivals	-0.0234	-0.0424	-0.0047

Weather: Open-Meteo Previous Runs provides GFS values predicted 24 hours before valid time. Fixed original XGBoost parameters, identical targets and identical simulation paths isolate the added inputs. Differences above are paired local-day MAE changes versus no weather. No target-hour observation or reanalysis replaces the forecast.

Duration: 30/60/90/120/180/360-minute cutoffs were recomputed from all original ZIPs for the selected network, with fixed model architectures and policy scenarios. The 180-minute endpoint counts and test errors exactly reproduce the baseline. Other cutoffs change the retained-record target, so lower errors alone do not prove a better model.

strategy	unmet	bike_km	service_proxy
greedy	1620	255.8964	0.9162
milp	1508	489.2008	0.9220
none	1752	0.0000	0.9094
stochastic_mpc	1436	782.9054	0.9257

Stochastic MPC: three-hour recursive forecasts exclude all future actual lags; four joint validation-residual-block scenarios preserve cross-station/temporal dependence. Shared integer moves and continuous scenario stocks optimize expected aggregate penalties plus terminal safety, with a requested 2% gap. Only the first move is executed each hour. All four policies use the same 48-hour event paths; solver/fallback status is logged. This remains a count approximation, not a latent-demand or vehicle dispatch model.

Road/service integration: a new 48-hour paired-event scenario starts at the first eligible measured historical state and uses a directed OSRM road proxy, two capacity-six co-loaded trucks, travel plus 90-second/bike handling, depot returns and driver bounds. All 181 routes and actual cargo ledgers reconcile. No-intervention service is 94.776%, road MILP 94.498% and road Greedy 94.420%; this unfavorable result is retained. Instant MILP is explicitly a resource relaxation. Roads are current driving proxies, capacity is frozen and the depot assumed. Full protocol/results are in operations_completion.md; the earlier restricted atomic audit remains a separate experiment.

Scale: a 60-station training-selected network has 525,600 panel rows and 87,900 test rows; all models and three policies execute, including 336 simulation hours. A version-26c official water-inclusive polygon verifies membership of the original 30 and defines this extension geography; it is not mislabeled as 2025 stock evidence. Keeping the bike budget at 12 tests scale/resource pressure on a different demand footprint.

Revised conclusion: historical evidence materially improves calibration but does not identify unrecorded operational states or rejected user requests. Raw polling heartbeats and flag-only/maintenance updates are still needed to remove the remaining coverage gap. The data-request draft has not been sent. New code/tests, route-service integration and collector deployment are complete within their declared scope; perfect historical truth and actual operator truck dispatch are not claimed.